TRIANGLE EQUIVALENCES OF DISTANCE VALUE SETS ACCUMULATING AT ZERO ARE SIMILARITIES

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Editorial note. The entire author proof is retained. At1028 the triangle estimate supplies <=f(delta_i)<epsilon, sufficient for continuity. At1043 an attained finite upper endpoint is covered by the already established continuity from below; the independent review explains this boundary instance of the author limit argument.

Editorial scope. One original rigidity theorem, including its assertion about every equivalence witness. Ordinary real analysis is allowed. Neither the author discrete-distance rigidity theorem nor the Borel order-coding theorem is a free premise or an additional counted item.

Allowed standard prerequisites (editorial).

- Elementary real analysis. Uniform continuity on intervals, density, rational approximation and limits.. Convert triangle constraints into continuity and then constant ratios; none of the rigidity conclusions is assumed.

Definition 0.1 ([1])**.** Let Δ be a nonempty set of positive real numbers. We call Δ a *distance value set* if for all $x, y \in \Delta$,

$$\min\{x + y, \sup \Delta\} \in \Delta.$$

Briefly speaking, a distance value set is exactly an initial segment of a subsemigroup of positive reals. Given a distance value set Δ, a metric space (X, d) is a Δ-*metric space* if $d(x, y) \in \Delta$ for all $x \neq y \in X$.

Definition 0.2 ([1])**.** Let Δ and Λ be distance value sets.

(i) A triple (x, y, z) of positive real numbers is a Δ-*triangle* if $x, y, z \in \Delta$ and $|x - y| \leq z \leq x + y$.
(ii) We say that Δ and Λ are *equivalent* if there is a bijection $f \colon \Delta \to \Lambda$ such that for any positive real numbers x, y, z, (x, y, z) is a Δ-triangle if and only if $(f(x), f(y), f(z))$ is a Λ-triangle.

Lemma 0.1. *$\approx_{\mathsf{CDVS}}$ is Borel reducible to an isomorphism relation of countable structures. Consequently, $\approx_{\mathsf{CDVS}}$ is Borel reducible to $E_{S_\infty}^{\infty}$.*

Proof. To each $\Delta \in \mathsf{CDVS}$ we associate a countable structure $\mathcal{A}_\Delta = (\Delta, R^{\mathcal{A}})$, where $R^{\mathcal{A}}$ is a ternary relation defined as

$$R^{\mathcal{A}}(d_1, d_2, d_3) \iff (d_1, d_2, d_3) \text{ is a } \Delta\text{-triangle.}$$

It is clear that for $\Delta, \Lambda \in \mathsf{CDVS}$, we have $\Delta \approx \Lambda$ if and only if $\mathcal{A}_\Delta \cong \mathcal{A}_\Lambda$. This gives a Borel reduction from $\approx_{\mathsf{CDVS}}$ to an isomorphism relation of countable structures. □

Theorem 0.2. *Let Δ and Λ be countable distance value sets with $\inf \Delta = \inf \Lambda = 0$. Then the following are equivalent:*

(1) *Δ and Λ are equivalent.*
(2) *There is a positive real number r such that $\Lambda = r \cdot \Delta$.*

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Moreover, if Δ and Λ are equivalent, then every witness of their equivalence has the form of multiplication by a positive real number.

Proof. The implication (2)⇒(1) is obvious. We only prove (1)⇒(2). For this, let f be a bijection between Δ and Λ witnessing their equivalence. Let $\mathcal{A}_\Delta = (\Delta, R^\Delta)$ and $\mathcal{A}_\Lambda$ be the countable structures defined in the proof of Lemma 0.1. Then f is an isomorphism witnessing $\mathcal{A}_\Delta \cong \mathcal{A}_\Lambda$. Since Δ is a distance value set and $\inf \Delta = 0$, we have that Δ is dense in $(0, \sup \Delta)$. Similarly, Λ is dense in $(0, \sup \Lambda)$.

Fix a sequence $(\delta_i)_{i\in\mathbb{N}}$ in Δ such that $\delta_{i+1} < \delta_i/2$ for all i. Such a sequence exists since $\mathcal{A}_\Delta$ satisfies the sentence $\forall x\, \exists y\, \neg R(y, y, x)$. Since f is an isomorphism, we have $f(\delta_{i+1}) < f(\delta_i)/2$ for all i.

We claim that f is continuous as a function from $\Delta \subseteq \mathbb{R}$ to $\Lambda \subseteq \mathbb{R}$. To see this, fix $x \in \Delta$. For any $\epsilon > 0$, find an i such that $\delta_i < x$ and $f(\delta_i) < \epsilon$. For any $y \in \Delta$ with $|x - y| < \delta_i$, (x, y, δ_i) is a Δ-triangle. Thus $(f(x), f(y), f(\delta_i))$ is a Λ-triangle. In particular, $|f(x) - f(y)| < f(\delta_i) < \epsilon$. The actually proves that for all positive δ, f is uniformly continuous on $\Delta_{>\delta} \triangleq \Delta \cap (\delta, +\infty)$.

Next we show that f is in fact linear. In the proof we use notation such as $\Delta_{>\delta}$, $\Delta_{<\delta}$, and $\Delta_{[a,b]}$ with obvious meanings.

To show that f is linear, assume $x, y, x + y \in \Delta$. We claim that

(a) $f(\Delta_{[|x-y|,x+y]}) \subseteq \Lambda_{[|f(x)-f(y)|,f(x)+f(y)]}$;
(b) $f(\Delta_{>x+y}) \subseteq \Lambda_{>f(x)+f(y)}$ or $f(\Delta_{>x+y}) \subseteq \Lambda_{<|f(x)-f(y)|}$;
(c) $f(\Delta_{<|x-y|}) \subseteq \Lambda_{>f(x)+f(y)}$ or $f(\Delta_{<|x-y|}) \subseteq \Lambda_{<|f(x)-f(y)|}$.

Let $z \in \Delta$. (a) is equivalent to saying that if (x, y, z) is a Δ-triangle, then $(f(x), f(y), f(z))$ is a Λ-triangle, which is immediate from our assumption. Also immediate is that if $z > x+y$, then (x, y, z) is not a Δ-triangle. Thus $(f(x), f(y), f(z))$ is not a Λ-triangle, and so either $f(z) > f(x) + f(y)$ or $f(z) < |f(x) - f(y)|$. Similarly for $z < |x - y|$. Now, to prove (b), assume toward a contradiction that there are $z_1, z_2 \in \Delta$ with $z_1, z_2 > x + y$ such that $f(z_1) < |f(x) - f(y)|$ and $f(z_2) > f(x) + f(y)$. Without loss of generality assume $z_1 < z_2$ (the argument for the case $z_1 > z_2$ is similar). By the continuity of f and the density of both Δ and Λ, we may find $z_1 = a_1, \ldots, a_n = z_2$ in Δ such that $a_k > x + y$ and $|f(a_{k+1}) - f(a_k)| < f(x) + f(y) - |f(x) - f(y)|$ for all $1 \le k \le n$. Then there must be an a_k with $|f(x) - f(y)| \le f(a_k) \le f(x) + f(y)$, contradicting $a_k > x + y$. This proves (b). A similar argument for $z_1, z_2 < |x - y|$ proves (c).

Since f is a bijection from Δ to Λ, the claimed inclusions are actually equalities. By considering sufficiently small δ_i, we see that $f(\Delta_{<|x-y|}) = \Lambda_{<|f(x)-f(y)|}$, and thus $f(\Delta_{>x+y}) = \Lambda_{>f(x)+f(y)}$. By the continuty of f again, we have $f(x + y) = f(x) + f(y)$.

Finally, we argue that for any $x, t \in \Delta$, we have $f(x)/x = f(t)/t$. This proves (2).

For each $k > 0$, find $0 < \delta_k < 1/2^k$ by the continuity of f such that for all $z \in \Delta$, $|t - z| < \delta_k$ implies $|f(t) - f(z)| < 1/k$. Now for each k, find positive rational number n_k/m_k such that $0 < t - (n_k/m_k)x < \delta_k/2$. Note that by the density of Δ and bijectivity of f, $\delta_k \to 0$ as $k \to \infty$. Thus $n_k/m_k \to t/x$ as $k \to \infty$.

Fix k. By the density of Δ, find an increasing sequence $(t_j)_{j\in\mathbb{N}}$ with $t_j \in \Delta$ and $t_j \to x/m_k$ as $j \to \infty$. Then $m_k t_j \to x$ as $j \to \infty$, and $m_k t_j < x \in \Delta$. Thus $m_k t_j \in \Delta$ and $m_k f(t_j) = f(m_k t_j) \to f(x)$, which gives $f(t_j) \to f(x)/m_k$. On the other hand, $n_k t_j < (n_k/m_k)x < t \in \Delta$ and $n_k t_j \to (n_k/m_k)x$ as $j \to \infty$. Hence $n_k t_j \in \Delta$ and $f(n_k t_j) = n_k f(t_j) \to (n_k/m_k)f(x)$ as $j \to \infty$. Thus we can find a j such that $|n_k t_j - (n_k/m_k)x| < \delta_k/2$ and $|f(n_k t_j) - (n_k/m_k)f(x)| < 1/k$. For this fixed j, we have $|t - n_k t_j| \le |t - (n_k/m_k)x| + |n_k t_j - (n_k/m_k)x| < \delta_k$. Thus $|f(t) - f(n_k t_j)| < 1/k$, and consequently $|f(t) - (n_k/m_k)f(x)| < 2/k$.

Letting $k \to \infty$, we get $f(t) = (t/x)f(x)$ and thus $f(x)/x = f(t)/t$. □

References

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